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Original Research

Impact of a Rule-Based Recommendation System on Female Students' Awareness of How to Search for Suitable Professional Certificates: A Quasi-Experimental Study Using Paired-Sample T-Test

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Abstract: This study examines the impact of an intelligent, rule-based recommendation system designed to address a key gap in literature; while most AI-based advising tools focus on course selection, few guide learners toward professional certifications that build career readiness. A quasi-experimental pre–post design was conducted with sixty-three female undergraduate students at a Saudi university, where limited awareness of certification pathways has been noted. Data were collected through a validated electronic questionnaire, and analyses were performed using Python for accuracy and transparency. Grounded in the Technology Acceptance Model (TAM), the study interprets awareness gains through students' engagement, motivation, and perceived usefulness of the system. Results from the paired-samples t-Test showed significant improvements in students' understanding of effective search strategies and their awareness of officially endorsed certificates and associated costs. Although the effect size was small (Cohen's d = 0.28), the findings indicate that even short exposure to an AI-driven tool can enhance professional awareness. Practically, the study highlights the potential of such systems to support academic advising and align higher education outcomes with labor market needs.

Keywords: Higher Education, Recommendation Systems, Professional Certificates, Student Awareness, Paired-Samples T-Test

Introduction

The digital transformations (DTs) witnessed globally in recent years have significantly reshaped labor market requirements, where technology and innovation now play a central role in defining job competencies. Preparing university students to navigate these rapid changes requires providing updated educational and training resources that align with modern workforce demands. Within this context, professional certifications have become an essential mechanism for validating students' skills and improving their competitiveness and opportunities for continuous professional development.

Despite their importance, many university students, particularly females in Saudi Arabia, often exhibit limited awareness of professional certification pathways and how to identify appropriate credentials aligned with their fields. In this study, awareness refers to students' ability to recognize, understand, and evaluate available certification options, their requirements, and career relevance. Limited awareness of such pathways reduces students' ability to make informed decisions and benefit fully from these certifications.

To address this issue, AI-based recommendation systems offer a promising solution. An AI-based recommendation system is defined as an intelligent tool that analyzes user information and generates personalized suggestions to help learners navigate large sets of professional options efficiently. Developing a certification-oriented system has strong potential to support students in identifying suitable certifications and aligning their choices with evolving labor market needs in Saudi Arabia. However, empirical evidence on the effectiveness of such systems in higher education remains limited.

Accordingly, this study aims to examine the impact of an intelligent, rule-based recommendation system on improving Saudi female university students' awareness of professional certification search and selection. Therefore, the central research question is: How can an AI-driven recommendation system effectively enhance female students' awareness of professional certifications and evolving labor market requirements? Answering this question will support future educational policies and contribute to strengthening the alignment between higher education outcomes and workforce expectations.

Literature Review

This section reviews prior studies related to professional certification, recommendation systems in education, and the role of AI in supporting students' decision-making processes.

Professional Certificates and Their Impact on University Students

Professional certificates represent a vital means of documenting the practical skills and specialized knowledge that individuals acquire beyond the traditional academic framework. These qualifications play a crucial role in directly enhancing graduates' readiness to enter the competitive job market. By integrating these professional certificates into university programs, educational institutions can significantly aid in bridging the existing gap between theoretical academic skills and the actual, practical requirements of the job market. This integration is essential for preparing students for the realities they will face as they transition from the classroom to the workplace (Bruguera et al. 2025). However, it is important to note that there exists a considerable lack of awareness among students, particularly among Saudi students, regarding the benefits and requirements of obtaining professional certificates. This gap emphasizes the urgent need to adopt and implement effective educational strategies designed to enhance students' awareness and understanding of these important qualifications (Alenezi et al. 2024). Students who actively pursue and obtain professional certificates during their university studies tend to demonstrate a greater sense of confidence during job interviews and enjoy considerably better employment opportunities upon graduation. Consequently, professional certificates not only validate the skills and knowledge that graduates have developed, but they also significantly enable them to enter specialized fields of work more rapidly than their peers who do not hold similar certifications. This advantage can often translate into a faster path to career advancement and personal satisfaction in one's chosen profession (Peón et al. 2023). Thus, investing in professional certifications while in university can provide a noteworthy edge in today's highly competitive job market.

Recommendation Systems (Recommender Systems) in Education

Educational environments have witnessed significant development in the use of recommendation systems to personalize content and guide learners toward educational paths that align with their interests and levels (Cha et al. 2024). Recommendation systems have been shown to be an effective tool for personalizing the learning experience based on user data and behavior, enhancing learner engagement, and achieving educational goals. Recommendation systems contribute to reducing the cognitive load resulting from an abundance of educational content by providing personalized and relevant suggestions (Aucancela et al. 2023). Students have demonstrated improved academic performance and satisfaction with the learning process when using advanced recommendation systems based on artificial intelligence (AI) and data analysis techniques (Lampropoulos 2023).

The Impact of Recommendation Systems on Decision-Making

Several studies have shown that recommendation systems play a pivotal role in supporting educational decision-making, especially in environments characterized by multiple educational options and paths. These systems help learners make informed decisions about the courses or certificates that are appropriate for them (Ain et al. 2025). Similarly, recommendation systems based on performance and interest data analysis reduce learner hesitation when making a decision and increase satisfaction with the decision made. Conversely, well-designed recommendations reduce frustration and improve the user experience in digital educational environments (Ma et al. 2024).

Using the T-Test to Measure Educational Impact

The paired-samples t-Test, which is also commonly referred to as the dependent samples T-test, stands out as one of the most frequently utilized statistical methods in a variety of studies that focus on assessing change that results from a particular intervention. This test provides researchers with the capability to examine the statistical significance of differences that exist between two related averages. These averages often pertain to performance measurements that are gathered before and after the implementation of a specific educational tool or resource (Tipton and Olsen 2018; Varnell et al. 2008). Utilizing this test effectively measures the degree of effectiveness associated with the use of interactive learning environments within educational settings. The results of such analyses have shown clear statistical significance, demonstrating a positive impact attributed to the intervention being studied. This type of analysis not only contributes valuable insights but also plays a crucial role in educational research. It is specifically aimed at thoroughly evaluating the influence of new technologies or targeted training programs that are introduced into educational contexts. The application of the paired-samples t-Test thus serves as a pivotal method in understanding how various interventions can affect learning outcomes and overall educational effectiveness (Suh 2025).

AI and Its Support for Educational Decision-Making

AI has increasingly emerged as an essential and transformative component in the development of various educational support systems, particularly in the crucial areas of recommendations and comprehensive learner data analysis. AI tools can significantly enhance the personalization of educational content through sophisticated analyses of learner behavior, preferences, and interests. This capability not only aids in customizing learning experiences but also plays an important role in improving the overall quality of educational decision-making processes (Ramanjaneyulu et al. 2024). The profound potential of AI lies in its ability to collect, process, and analyze vast amounts of data regarding student performance. Such data can subsequently be utilized to generate accurate and tailored recommendations that bolster self-directed learning and informed decision-making. Furthermore, AI applications within the realm of education are instrumental in predicting potential academic failure, allowing for timely interventions and improved teaching strategies that are informed by real-time data. Through these capabilities, AI continues to revolutionize the educational landscape, fostering a more data-driven and responsive approach to teaching and learning (Shwedeh 2024).

Research Gap

Despite growing interest in AI-enabled guidance tools in higher education, most recommended systems focus on course selection or content personalization rather than steering learners toward industry-recognized professional certifications that build career readiness. Recent evaluations of AI-based course recommenders highlight efficiencies in information search and decision support, yet they rarely extend to certification pathways or explicitly report outcomes such as awareness of professional credentials (Cha et al. 2024). At the same time, research on women undergraduates in Saudi Arabia underscores distinctive contextual needs and evolving career intentions, suggesting that generic guidance tools may overlook gender- and locale-specific factors shaping decision-making (McNally and Winkel 2023). Accordingly, prior work leaves two gaps: (1) the absence of recommender systems tailored to professional certification choices and (2) the lack of evidence on such systems' effects among Saudi female students. Our study addresses these gaps by designing and testing a professional-certification recommender targeted to Saudi female undergraduates and evaluating its impact on certification awareness.

Conceptual Framework

Figure 1 illustrates the conceptual model underlying this study. The framework proposes that the AI-based recommendation system influences students' awareness indirectly through three mediating variables: engagement, motivation, and perceived usefulness. Recent research has shown that AI-supported learning environments enhance students' internal motivation and overall learning experience by providing adaptive and personalized interactions (Mohamed et al. 2025). Additionally, perceived usefulness has been identified as a central factor influencing learners' acceptance and intentional use of AI tools in educational contexts (Wang et al. 2023), consistent with the Technology Acceptance Model (TAM). Moreover, students' engagement and meaningful interaction with AI tools have been found to shape the perceived relevance and value of the learning experience, reinforcing awareness-building and reflective decision-making (Chan and Hu 2023). Based on these assumptions, the model theorizes that intelligent recommendation systems enhance awareness not directly, but by supporting engagement, motivation, and perceived usefulness within the learning process.

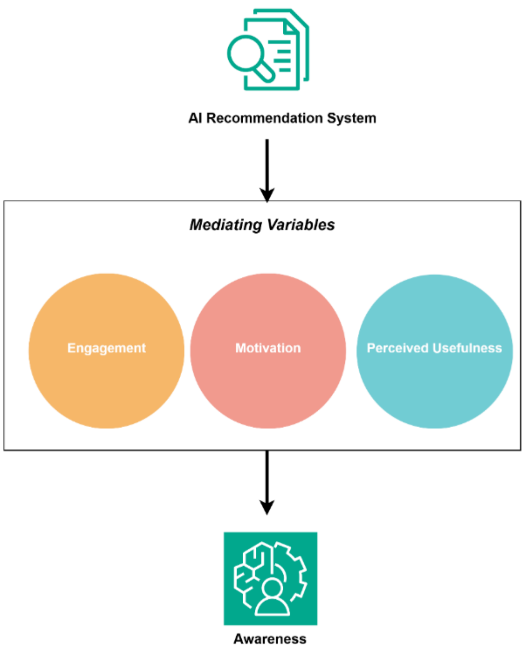

Figure 1: Conceptual Model

Methodology

Study Design

The experimental study was designed to measure the effectiveness of the professional certification recommendation system in increasing students' awareness of how to search for these certificates and their role in improving their skills and career opportunities. The experimental design relied on a before-and-after approach, measuring students' awareness before and after using the recommendation system. Questionnaires were used to measure students' awareness before and after using the recommendation system, and responses were analyzed using appropriate statistical methods to determine the extent of the impact. The study was designed to collect data on students' awareness of how to search for professional certificates. The study was conducted in two phases: the first, before the implementation of the rule-based recommendation system, aimed to measure students' initial awareness; the second, conducted after the pilot, measured changes in awareness resulting from the system's use. The study focused on several key aspects related to professional certificates, such as knowledge of professional certificates in the field of specialization and how to search for them. A variety of questions were used to collect quantitative data that could be analyzed statistically. To determine differences in results, a paired-samples t-Test was appropriately used on the responses of the same group in both phases. The study was piloted in computer laboratories under the researcher's supervision, adhering to scientific standards to maintain the reliability of the results. Notably, the system does not require students to create an account or provide any personal information. It is simply a tool to help search for professional certificates that fit the specializations and needs of undergraduate students, enhancing the concepts of data protection, flexibility, and ease of use. Responses were collected in a structured environment. The study results demonstrated a significant increase in students' awareness of how to search for professional certificates after implementation, indicating that rule-based systems can effectively contribute to enhancing students' skills and awareness of these certificates, in line with the principles of statistical analysis and previous research.

Instrument Validity and Reliability

To ensure the instrument's soundness and accuracy, two complementary procedures were undertaken to establish its validity and reliability. First, the questionnaire was thoroughly reviewed and approved by the academic supervisor, who evaluated the alignment of each item with the research objectives and theoretical constructs. This process ensured content validity, confirming that the items were conceptually relevant, clearly stated, and covered the dimensions intended to measure students' awareness of professional certifications. Such supervisory validation aligns with the recommendations of Romero Jeldres et al. (2023), who emphasized that expert or supervisory review can serve as an adequate procedure for ensuring content validity in educational and social research instruments.

Second, to verify the internal consistency of the instrument, Cronbach's Alpha was calculated using Python, resulting in a value of 0.727. According to Hussey et al. (2025), values of $\alpha \geq 0.70$ are considered acceptable indicators of internal consistency, particularly in exploratory studies. This finding is consistent with Pérez-Rivas et al. (2023), who reported similar reliability coefficients ($\alpha \approx 0.72$) as satisfactory in validating a newly developed instrument. Therefore, the obtained value indicates that the questionnaire items were adequately homogeneous and reliable for measuring the construct of students' awareness. Overall, both the academic review and the statistical reliability analysis confirm that the instrument used in this study achieved an acceptable level of validity and reliability, providing a strong foundation for the interpretation of results.

Justification for Using a Paired-Samples T-Test

The choice of a paired-samples t-Test in this study stems from the need for an accurate comparison between student awareness measurements before and after the implementation of a rules-based professional certification recommendation system. This test is appropriate because the collected data relates to two related measures for the same group of students, whose awareness levels were measured before and after the system was implemented. The importance of using this test lies in its ability to determine whether changes in students' awareness are statistically significant, which directly reflects the effectiveness of the system in improving their awareness. Furthermore, the paired-samples t-Test allows for the analysis of small differences between the two related groups, due to its high sensitivity, ensuring that the results derived reflect the true change resulting from the intervention. Data analysis is straightforward and efficient, and most importantly, the data are based on identical measurements for the same individuals, reducing the potential for intervening variables to influence the results. Based on the limitations of this study, including the nature of the data and the study design, a paired-samples t-Test provides the optimal means for accurately and reliably measuring the system's impact. This strengthens the validity of the results and, consequently, the study's conclusions regarding the system's effectiveness in raising students' awareness of how to search for professional certificates.

Implementation Environment

The system was implemented on female students. The study sample included a group of sixty-three female students from the College of Business at King Khalid University. They were randomly selected to ensure fair representation through practical lectures in computer laboratories. A group of female students expected to graduate were selected as the target group for this study. During the first session, students were asked to complete a pre-questionnaire to measure initial awareness. This was followed by an introductory session to explain how to use the recommendation system effectively and to provide detailed explanations on how to utilize it in research and selecting appropriate professional certificates for each student. In the second session, the system was implemented to track certification selection and assess students' awareness of the recommendations provided. Students were also asked to complete a post-questionnaire to analyze the results. Continuous technical support was also provided to ensure that necessary assistance was available throughout the implementation period. The environment was designed to be interactive, allowing students to interact effectively with the system. All activities were regularly monitored, and relevant responses were documented to ensure the accuracy and transparency of the results. This careful planning of the environment and the ease of access to the system provided ideal conditions for measuring the system's impact and achieving reliable and verifiable results, highlighting the importance of the applicable environment in ensuring the successful use of the recommendation system by students.

Method of Data Analysis

When analyzing the survey results, it should be clearly noted that the analysis method used relies primarily on the Google Colab tool and the Python programming language. This approach ensures high accuracy while minimizing potential analytical errors. The statistical results showed that the implementation of the rule-based professional certification recommendation system led to a clear increase in students' awareness of how to search for professional certificates. The average awareness scores after implementing the system increased compared to previous periods. The results also indicated that the differences between the averages before and after implementation were highly statistically significant, reflecting the system's effectiveness in enhancing the knowledge and awareness of the targeted students. From an educational perspective, these results can be interpreted as meaning that the system effectively contributed to improving the way students search for professional certificates that fit their needs and specializations, which led to greater focus on making sound decisions and providing clear insights, rather than distraction and wasting time. Additionally, the results indicated a significant improvement in students' perceptions of the value of certificates and their role in building successful career paths. This reflects a positive impact on their behavior and attitudes toward pursuing these certificates, which can lead to increased employment opportunities and professional development. This is consistent with previous studies that have emphasized the role of AI-based guidance systems in raising awareness and motivating students to engage in continuing education programs. It is worth noting that the results underscore the ongoing need to develop and expand the scope of these systems to ensure they reach the largest number of students, while also taking into account the diversity of individual needs and designing more accurate measurement tools to track future developments.

Assumption Testing

Before conducting the paired-samples t-Test, the differences between pre- and posttest scores were examined for normality using the Shapiro–Wilk test ($p = 0.171 > 0.05$), indicating no significant deviation from normality. Outlier analysis revealed five outliers; however, their proportion was small relative to the total sample size ($n = 63$), so they were retained for analysis. Given the presence of outliers, a Wilcoxon signed-rank test was also performed, which confirmed the significance of the difference ($W = 497.5$, $p = 0.022$). Additionally, the boxplot and Q–Q plot further supported the assumption of an approximately normal distribution of the differences.

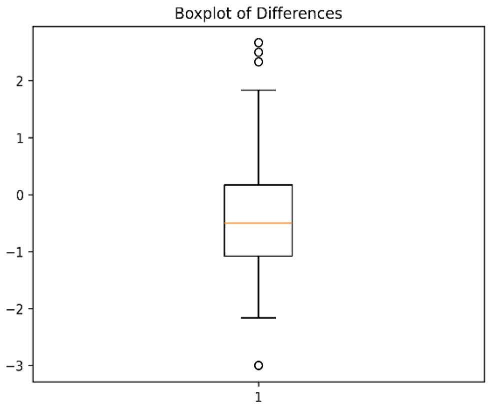

Figure 2: Boxplot of Data Sample

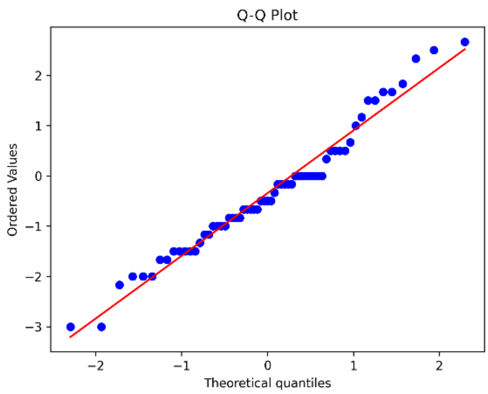

Figure 3: Q–Q Plot for Normality Assessment

Statistical Results

The results of the paired-samples t-Test for the overall mean score (all items combined) indicated a statistically significant difference between the pretest ($M = 2.183$) and posttest ($M = 1.839$) scores, $t(62) = 2.206$, $p = 0.031$, with a small effect size (Cohen's $d = 0.28$).

Table 1: Paired Samples T-Test for Overall Awareness Scores

Participants N	*Mean Pre*	*Mean Post*	*Difference (Pre–Post)*	*T*	*P*	*Effect Size (Cohen's d)*	*Interpretation*
63	2.183	1.839	0.344	2.206	0.031	0.28	Small effect

The results showed statistically significant differences in two main axes of the questionnaire, reflecting the positive impact of using the recommendation system on students' awareness. In the applied Likert scale, lower mean values indicate higher levels of agreement and awareness.

Axis I: Knowledge of Searching for Appropriate Professional Certifications

The phrase "I know how to search for the appropriate professional certificate for me" represents this axis. The results of the T-test showed a statistically significant difference between the pre- and posttest averages, with the T-value reaching 4.119 and a significance level of $P = 0.0001$, indicating a tangible improvement in the students' ability to search for certificates after using the system. The arithmetic averages also showed a decrease in the degree of agreement with the statement after the experiment, indicating an increase in the participants' awareness. These averages are summarized in an appendix to the scientific paper, showing the differences between the pre- and posttest responses for each statement, with a reference to the T-test results that support or do not support the statistical significance of these differences. This numerical representation is a crucial step in verifying the study's hypothesis, as it allows for accurate and transparent presentation of the differences.

Table 1: The Paired-Samples T-Test Results

No.	*Phrase*	*T-Value*	*P-Value*	*Mean (Before)*	*Mean (After)*	*Effect Size (Cohen's d)*	*Significant Difference*
1	I know which certificates are appropriate for my university major.	1.578	0.1214	2.08	1.76	0.20	There is no significant difference
2	I have an idea about the certificates programs supported by Human Resources Development Fund (HADAF) and their costs.	2.483	0.0158	2.32	1.76	0.31	There is a significant difference
3	I know the difference between the types of professional certificates (technical, administrative, etc.).	1.210	0.2311	2.16	1.89	0.15	There is no significant difference
4	I feel confident in my ability to choose an appropriate professional certificate.	1.000	0.3212	1.97	1.78	0.13	There is no significant difference
5	I have a clear career goal I am striving to achieve.	1.154	0.2530	1.75	1.97	0.15	There is no significant difference
6	I know how to find the right professional certificate for myself.	4.119	0.0001	2.83	1.87	0.52	There is extreme significant difference

Axis II: Awareness of Supporting Agencies and Certification Costs

This axis was represented by the phrase: "I have an idea about the professional certificates supported by the Human Resources Development Fund (HRDF) and their costs." The results also showed a statistically significant difference, with the T-value reaching 2.483 and a significance level of $P = 0.0158$. This indicates that the system contributed to enhancing students' awareness of the entities that provide financial support for degrees, which is one of the primary objectives the system sought to achieve.

Other Axes

The remaining axes, such as feeling confident in choosing the appropriate degree, clarity of career goals, and knowledge of the types of certificates, showed slight changes in the averages. However, the T-test did not reveal statistically significant differences ($P > 0.05$). Nevertheless, the general trend of the responses indicates a relative improvement in students' awareness, reflecting the system's role as an initial catalyst for considering career options, even if the effect was not statistically significant across all axes.

Discussion

The findings of this quasi-experimental study indicate a statistically significant improvement in students' awareness of professional certifications after interacting with the AI-based recommendation system. Although the effect size was small (Cohen's $d = 0.28$), the result provides meaningful evidence that even limited exposure to an intelligent recommendation tool can enhance students' understanding of professional learning pathways. This improvement in awareness can be explained through three interrelated mechanisms: interaction, perceived ease of use, and motivation. First, the interactive design of the system encouraged active engagement, allowing learners to explore personalized recommendations rather than receiving static information. Such interaction likely promoted deeper cognitive processing and reflection on career-related decisions. Second, the system's ease of use reduced cognitive barriers, aligning with the TAM, which emphasizes that perceived usefulness and perceived ease of use influence users' attitudes and behavioral intentions. Third, the personalized nature of the recommendations may have served as a motivational factor, increasing students' curiosity and sense of relevance toward professional certificates aligned with their fields of study.

The small effect size ($d = 0.28$) may be attributed to the short duration of the intervention and the limited sample size. Since the exposure lasted only a few sessions, participants may not have had sufficient time to fully explore the system's features or reflect on their professional goals. Additionally, the relatively small and homogeneous sample could have constrained the variability in awareness scores, thereby reducing the observed effect magnitude.

These results are consistent with prior research highlighting the positive, though sometimes modest, impact of AI-driven learning tools on students' motivation and awareness. For instance, Cha et al. (2024) reported that AI recommendation systems improved learners' engagement and perceived usefulness but required longer-term integration to yield larger cognitive gains. Similarly, Lampropoulos (2023) found that adaptive educational technologies enhance learners' awareness and goal orientation, particularly when combined with interactive feedback and user control. Together, these findings reinforce the notion that short-term interventions can initiate awareness improvements, while sustained interaction and iterative feedback are necessary to achieve more substantial educational outcomes. However, not all awareness statements demonstrated a statistically significant improvement. This can be explained by the cumulative nature of professional awareness, which typically develops through repeated exposure, real-world experience, and progressive goal clarification. Some items in the questionnaire required deeper understanding of certification pathways, career planning, or institutional processes knowledge that typically emerges over extended engagement rather than a short-term intervention. This aligns with the idea that awareness is a cumulative construct shaped by sustained cognitive engagement, perceived task relevance, and meaningful usability experiences. Therefore, the observed partial gains are expected at this early stage and suggest that longer or repeated interaction with the system may yield stronger effects.

Implications and Applications

This study provides several practical implications for educational policymakers and university administrators. Saudi universities can integrate intelligent recommendation systems such as *Ehtirafi* into their career guidance and academic advising centers to enhance students' access to relevant professional certifications aligned with their academic majors and labor market needs. Implementing such systems supports national strategies under Vision 2030, particularly in developing human capital, promoting employability, and advancing DT within higher education.

At the policy level, universities could consider embedding AI-driven recommendation tools within their institutional career centers, supported by guidelines for data governance, ethical use, and system evaluation. Establishing formal partnerships between universities and certification providers would also ensure the alignment of recommended credentials with current industry requirements. A practical implementation plan could involve three phases: (1) piloting the system in business and management programs, (2) training academic advisors on its use, and (3) gradually expanding its deployment across faculties once institutional validation is achieved.

Furthermore, future iterations of *Ehtirafi* could incorporate adaptive learning algorithms and predictive analytics to dynamically personalize recommendations based on user profiles, feedback, and evolving career trends. Integrating multilingual interfaces and cross-platform accessibility could further enhance inclusivity and user engagement, particularly for diverse student populations.

While the current findings are contextually grounded in Saudi Arabia, they hold potential for broader applicability. Comparative studies across international contexts could examine how cultural and gender dynamics influence the adoption and perceived usefulness of AI-based career guidance systems. Such cross-contextual investigations would not only enhance the generalizability of this research but also position *Ehtirafi* as a model for global higher education institutions seeking to digitalize their academic advising and career guidance services.

Conclusion

Based on the study's findings and observations from the practical application and research experience, it can be concluded that the use of an intelligent recommendation system for selecting professional certificates was an effective tool in enhancing female university students' awareness of their career options. The quantitative results demonstrated that the system significantly enhanced students' knowledge of effective certificate-search strategies and increased their awareness of supporting institutions and associated costs, two essential elements in students' career decision-making.

Through the actual implementation experience, the importance of providing customized technical tools that meet students' needs and provide them with organized, easy-to-understand information, enabling them to discover available opportunities based on personal criteria such as university major and other needs, emerged. The experiment also highlighted the significant role of intelligent technologies, such as recommendation systems, in supporting educational decision-making and bridging the gap between higher education outcomes and labor market needs.

Despite the positive results demonstrated by the study, some areas did not record statistically significant differences, indicating that raising professional awareness is a cumulative process that cannot be reduced to a single short experience. Enhancing this awareness requires a long-term strategic approach, including the continuous and gradual integration of such systems into educational settings, in addition to providing accompanying human guidance.

In conclusion, this study emphasizes the promising potential of technology in improving higher education outcomes and its role in supporting university students to become more aware, confident, and independent in building their professional future. By expanding the adoption and continuous development of such systems, educational institutions can effectively contribute to preparing a generation more aligned with the demands of the labor market.

Recommendations

Considering the above, the study proposes the following recommendations:

- Promote the adoption of educational recommendation systems in Saudi universities and other educational institutions, particularly those targeting career and academic guidance, while adapting them to take into account the specificities of each major and academic stage.
- Integrate the recommendation system into the university advising services available to students so that the system does not replace the academic advisor but rather supports them, thereby enhancing the quality of recommendations received by students.
- Expand the scope of the system to include additional features such as personality analysis or career inclinations, which enables the provision of more personalized and accurate recommendations and contributes to more informed career decisions.
- Conduct longitudinal studies to measure the impact of using the system over a longer period of time, to verify and enhance its impact, and to monitor any changes that may occur in students' awareness and career orientations.
- Conduct comparative studies between students using the system and those relying solely on traditional guidance to measure differences in awareness and decision-making levels and to determine the extent of integration between the two methods.
- Provide training workshops for students and academic advisors to enable them to use recommendation systems efficiently and maximize their benefits, achieving integration between human and technical guidance.
- Ensure that the system database is constantly updated; ensuring it includes the latest professional certificates, requirements, and available support; ensuring the reliability and accuracy of the information provided.
- The implementation can follow a structured roadmap consisting of (1) system integration with existing LMS platforms, (2) training advisors using standardized operational guidelines, and (3) continuous monitoring and iterative refinement based on user analytics and feedback.

Limitations and Future Work

While the recommendation system *Ehtirafi* yielded promising outcomes, this study acknowledges several limitations. First, the sample was relatively small, comprising only sixty-three female students from a single college, which restricts the generalizability of the findings across broader university populations and genders. Second, the intervention period was short-term, which may partially account for the modest effect size observed. Raising awareness of professional certifications is a cumulative process that benefits from repeated engagement, extended exposure, and continuous reinforcement. Therefore, the current results should be viewed as preliminary evidence of effectiveness. Future research is encouraged to employ longitudinal designs that assess the persistence of awareness gains over time and to conduct comparative studies involving both male and female students across multiple disciplines.

Expanding the participant pool and incorporating behavioral or career outcome measures would also enhance the external validity and practical relevance of subsequent findings.

AI Acknowledgment

I acknowledge the use of ChatGPT (OpenAI) to support translation and improve the clarity, coherence, and academic tone of the manuscript. The prompts used included:

- Requests for translating academic content from Arabic to English.
- Rephrasing sentences for academic clarity and conciseness.
- Improving grammatical accuracy and academic style.
- Enhancing paragraph structure and transitions.

The output from these prompts was used solely for language refinement and structural organization. All AI-generated content was carefully reviewed, edited, and verified by the author to ensure accuracy, originality, and compliance with academic standards. While the author acknowledges the usage of AI, the author maintains that she is the sole author of this article and takes full responsibility for the content therein, as outlined in COPE recommendations.

Informed Consent

The author has obtained informed consent from all participants.

Conflict of Interest

The author declares that there is no conflict of interest.

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